

Neeraj Balachandar



Education

Indian Institute of Technology (IIT) Hyderabad

Dual Degree: *B.Tech in Mechanical Engineering* — *M.Tech in Aerospace Engineering* — Minor in **Robotics**

Hyderabad, India

Nov 2022 – Present

CGPA: 3.5/4

Relevant Coursework

- **Mathematics:** Matrix Theory, Linear Algebra, Probability and Stochastic Processes, Information Theory
- **Dynamics & Control:** Control Systems, Optimal Control, Convex Optimisation, Spacecraft Dynamics and Control
- **Computational Mechanics:** Incompressible Fluid Flows, CFD, FEM, Scientific Parallel Computing
- **Robotics:** Autonomous Robotics, Soft Robotics
- **Learning & Decision Making:** Machine Learning, Reinforcement Learning, Multi-Agent Systems, Principles of Game Theory

Research Contributions

Journal Publications

- **Neeraj Balachandar**, Yashwanth M., Vishnu R. Unni. "A Geometry-Informed, Multi-Task Neural Operator for Nonlocal Kernel Interactions in Lagrangian Flows." *Manuscript in preparation; preprint forthcoming on arXiv; [GitHub](#)*
- **Neeraj Balachandar**, A. Padmaprabhan, Vishnu R. Unni. "An Aeroelastic Solver Integrating Reformulated-Vortex-Particle and Finite-Element Methods across Non-Conforming Interfaces". *Manuscript in preparation for submission to Elsevier; preprint on arXiv; [GitHub](#)*
- Yashwanth M., **Neeraj Balachandar**, Dhinesan, Harishankar M, Akhin, Vishnu R. Unni. "Modeling and Control of Aerial Robots from Constrained Dynamics" *Manuscript in preparation; preprint forthcoming on arXiv; [GitHub](#)*

Conferences

- **Neeraj Balachandar**, Shriram Hari, Vishnu R. Unni. "Nonlinear Model Predictive Control via Sequential Convex Programming for Drone-to-Drone Docking. [🔗](#)" *Presented at the IEEE/ASME International Conference on Advanced Intelligent Mechatronics (AIM 2026), July 10, 2026, Genova, Italy (In-Person Contributed Paper Session); [GitHub](#)*
- **Neeraj Balachandar**, Yashwanth M., and Vishnu R. Unni. "Physics Informed Fourier Neural Operator to Map Reformulated Vortex Particle Field to Eulerian Velocity Field." *Accepted for presentation at the 14th International Symposium on Turbulence and Shear Flow Phenomena (TSFP14), July 2026, Heidelberg, Germany, (Submission withdrawn by authors).*
- **Neeraj Balachandar**, Yashwanth M., Akash M., Mahathi Kesavan, and Vishnu R. Unni. "Actuator System for Directional Manoeuvre of a Flapping Wing Aerial Vehicle. [🔗](#)" *Presented at the 2025 AIAA SciTech Forum, Session AS-08: Bio-inspired Adaptive Structures, January 8, 2025, Orlando, FL (In-Person Technical Paper Session).*
- **Neeraj Balachandar**, Vishnu R. Unni. "Analytical Modeling of Deformable Membranes in Undulatory Motion driven by Tendon-based Cosserat Rods" *In preparation for submission to 10th IEEE International Conference on Soft Robotics.*

Thesis

- "Control-Aware Model Reduction and Morphology-Control Co-Design for Soft Dynamical Systems". *M.Tech thesis 2026-27*

Achievements

- Awarded the **Undergraduate Research Excellence Award**, given annually to **one student** in the Dept. of Mechanical and Aerospace Engineering at IIT Hyderabad for outstanding research contributions.
- Ranked **second among Indian teams** in the **Swarm Rescue Challenge 2025** [🔗](#), organized by CIEDS–École Polytechnique de Paris.
- Awarded the **Summer Research Fellowship** by the **Indian Academy of Sciences** for a research internship at the **Indian Institute of Science** in Summer 2024.
- Ranked in the **top 0.3 percentile** nationally in the **JEE 2022**.

Ongoing Research

Control-Aware Model Reduction and Morphology-Control Co-Design for Soft Dynamical Systems

Aug 2026- Present

Dr. Vishnu R. Unni [🔗](#), *Dr. Vishal Sawant* [🔗](#)

- Developing a **soft-robot simulation and action-state RL-Gymnasium** for a generalized class of soft robots to learn locomotion.
- Investigating **morphology-dependent controllability**, relating changes in the robot's dynamics and actuation to its achievable locomotion behaviors.
- Developing a **reduced-order representation of body-wake dynamics** that preserves locomotion- and control-relevant behavior.
- Extending the framework toward **morphology-control co-design**, where learned locomotion performance guides the evolution of the robot's body and control strategy.

A Geometry-Informed, Multi-Task Neural Operator for Nonlocal Kernel Interactions in Lagrangian Flows: DD-MT-GINO [🔗](#).

Sep 2025- Present

Dr. Vishnu R. Unni [🔗](#), *Dr. Sumohana S. Channappayya* [🔗](#)

- Formulated and implemented an operator learning framework, modifying the existing Geometry-Informed Neural Operator to perform **multi-task** prediction in a **shared Fourier latent space**.
- The architecture was validated by learning two coupled tasks from a common latent space; **vortex-particle evolution** in the Lagrangian frame and **Eulerian velocity-field reconstruction**, exploiting their shared underlying Poisson PDE dynamics, through an operator-theoretic formulation.
- The architecture is compared against the classical Fast Multipole Method as a **data-driven surrogate** for speedup while maintaining accuracy. Systematically studied input-output representations through parameter sweeps.
- Generalization to unseen kinematics, geometries, and efforts to achieve super-resolution for reconstruction are underway, along with attempts to make the architecture **abstract**, towards a more general multi-task neural operator framework.

An Aeroelastic Solver Integrating Reformulated-Vortex-Particle and Finite-Element Methods across Non-Conforming Interfaces: VarFLEXI [🔗](#).

May 2025 - Present

Dr. Vishnu R. Unni [🔗](#), *Dr. Niranjan Ghaisas* [🔗](#), *Dr. Prakhar Gupta* [🔗](#)

- We developed the framework VarFLEXI, a variable-fidelity partitioned solver that uses a meshless scheme to solve the Navier-Stokes equations with a **reformulated vortex particle method**, coupled with the FEniCS to solve dynamic aeroelasticity problems using *Generalized- α* .
- The solver features strong work-conservation techniques that exploit the **Common Refinement Method** (CRM) or **Wendland Kernels-based interpolation** (RBF), to transfer aerodynamic force and structural geometry across **non-conforming** and **multi-representative discretizations**, through a common interface.
- We obtained **close validation** against an experimental water-tunnel heaving-wing setup and performed multiple **convergence tests** for spatial and temporal discretization and wake fidelity. We also show the advantage of using **CRM over RBF** methods in reducing work-conservation error at the interface.
- Designed the framework primarily for **vortical-structure-dominated** flows, with variable-fidelity configurations that enable substantially lower computational cost than fully meshed solvers, reducing simulation times from **days to hours** while retaining essential bi-directionally coupled aeroelastic dynamics.

Nonlinear Model Predictive Control via Sequential Convex Programming for Drone-to-Drone Docking: [NMPC-SCP](#)

May 2025 - Present

Dr. Vishnu R. Unni

- Developed a **receding-horizon nonlinear model predictive control framework** for autonomous drone-to-drone docking, formulating docking as a finite-horizon constrained optimal-control problem and solving it through **Sequential Convex Programming (SCP)**.
- Formulated a **reduced-order nonlinear multirotor model augmented with wind-disturbance states**, and iteratively linearized the dynamics within a trust-region SCP framework with virtual control to obtain tractable convex subproblems.
- Integrated **disturbance-aware target-state estimation and prediction using a Kalman filter** with receding-horizon replanning, enabling trajectory generation under noisy measurements and target drift for both stationary and constant-velocity target motion.
- Developed a unified set of **kinematic, geometric, and safety constraints**, including docking-cone capture geometry, chaser-target collision avoidance, and actuation constraints, with nonconvex constraints convexified into **second-order cone programs**.
- Validated the planned trajectories in a **high-fidelity full rigid-body MuJoCo simulation** with an NMPC planner and high-frequency inner-loop attitude control; convex subproblems achieved approximately **12-18 ms** solve times per iteration, with negligible docking-cone violations, satisfying docking constraints.

Modeling and Control of Aerial Robots from Constrained Dynamics:

Jan 2026 - Present

[Learn-To-Fly](#)

Dr. Vishnu R. Unni

- Developed a mechanically constrained aerial-robot platform for **system identification and control**, using reduced translational and rotational degrees of freedom to obtain direct measurement of generalized coordinates and net wrench on the multi-rotor.
- Formulated the coupled aerial-robot dynamics through an **Euler-Lagrange framework**, deriving the mass matrix, Coriolis terms, potential-energy terms, and generalized forces for the constrained 3/4/5-DOF configurations.
- Developing a framework for **recovering the net wrench** acting on the aerial robot from generalized-coordinate measurements, with analysis of **observability, matrix rank, conditioning**, and sensor configuration across the attainable workspace.
- Learning **free-flight aerial-robot dynamics from constrained experimental data** using inverse-dynamics-based wrench estimation and data-driven models (**Koopman**), while systematically evaluating sensitivity to model-parameter uncertainty and sensor noise.

Experience

Undergraduate Researcher, Dynamics and Control Lab (DysCo), IITH

Hyderabad, India

Supervised by [Dr Vishnu R. Unni](#)

Mar 2024 - Present

- Conduct research on the **modeling and control of complex dynamical systems**, spanning **fluid-structure interaction, aerial robotics, and soft robotics**.
- Develop **data-driven models and reduced-order representations** for prediction, system identification, and control of nonlinear dynamical systems.
- Design **optimization and learning-based safe control** methods for constrained dynamical systems.
- Develop **reproducible scientific-computing frameworks in Julia and Python**, combining theory, numerical simulation, and **experimental/hardware-in-the-loop validation** for multiphysics and robotic systems.
- Teaching Assistant for **Spacecraft Dynamics and Control** (Spring 2026 - [Dr. Vishnu R. Unni](#)) and **Control Systems** (Fall 2026 - [Dr. Safvan Palathingal](#)); prepared assessment and assignment problems and evaluated student solutions.

[Swarm Rescue Challenge](#), CIEDS-Ecole Polytechnique de Paris

Palaiseau, France (hybrid)

Supervised by [Dr David Filliat](#)

Nov 2024 - Mar 2025

- Developed a **Python-based multi-UAV exploration and rescue framework** for unknown environments under **obstacle, communication, and resource constraints**.

- Designed decentralized swarm navigation using **Artificial Potential Fields** for exploration and **Rapidly-exploring Random Trees (RRT)** for return-to-base transport, with nearby drones maintaining a **coordinated polygon formation around a designated leader**.
- Evaluated swarm performance across **4 simulated environments** and compared clustering-based exploration strategies using **DBSCAN and K-means**; achieved an average of **4.5 rescues and 1.5 collisions per 2000 simulation steps**.
- Presented the system and challenge results **in person at École Polytechnique de Paris**; ranked **second among Indian teams** based on combined technical performance and presentation.

Research Intern, Indian Institute of Science

Supervised by [Dr SN Omkar](#)

Bangalore, India

May 2024 - Aug 2024

- Investigated **human neuromuscular response under near-microgravity loading** using experimental **EMG time-series and motion-capture data**, with relevance to wearable and assistive robotics.
- Designed and developed a **wearable exoskeleton to reproduce near-microgravity loading conditions**, integrating mechanical design with biomechanical and musculoskeletal modeling.
- Used **OpenSim** with experimental motion-capture data to validate musculoskeletal simulations and analyzed **elbow-flexor muscle activation from EMG** under altered loading conditions.

Selected Projects

Convex Synthesis of Control Barrier Functions: A Review of Safety-Critical Control

Mar 2026

[Dr. Sumohana S. Channappayya](#)

- Investigated **Control Barrier Function (CBF) synthesis under actuator constraints**, comparing classical CBFs, **Input-Constrained CBFs (ICCBFs)**, and Sum-of-Squares (SOS)-based formulations for safety-critical obstacle avoidance.
- Implemented and analyzed **polynomial barrier-function synthesis**, including joint barrier-controller parameterization; identified the resulting **bilinear matrix inequality (BMI)** and demonstrated how actuator saturation can invalidate safety guarantees obtained without explicit input constraints.
- Developed a **convex SOS/semidefinite-programming formulation** for input-constrained CBF synthesis using lifted polynomial representations, SOS multipliers, and Schur-complement-based input constraints; implemented and evaluated formulations using YALMIP, MOSEK, and Clarabel.

A Neuro-Fuzzy logic for PID control

Nov 2025

[Dr. Vishnu R. Unni](#)

- Developed an **adaptive neuro-fuzzy PID controller** for nonlinear trajectory tracking of a **2-DOF robotic manipulator**. Used **online gradient-based learning** to adapt PID gains and fuzzy membership parameters from joint tracking error and its rate of change.
- Evaluated tracking and gain adaptation under **parabolic, high-slope sinusoidal, and multi-frequency trajectories**, demonstrating better performance over fixed-gain PID.

Parallelization of SVD Algorithm

Feb 2025

[Dr. Niranjan Ghaisas](#)

- Implemented a **Jacobi-based SVD/eigendecomposition algorithm in C with OpenMP**, parallelizing matrix operations and off-diagonal pivot search for multicore execution.
- Benchmarked **serial vs. parallel performance and strong/weak scaling** across increasing matrix sizes.

Technical Skills

Programming Languages: Python, Julia, MATLAB, C, Embedded C

Tools: ROS, Gazebo, MuJoCo, Paraview, Simulink, SolidWorks, Ansys Fluent, Simulia Abaqus, LaTeX, Github, OpenSim, Linux OS, VS Code, QP solvers

Extracurricular activities

Part of the **University Basketball team** (Inter IIT 2023-2025)

Completed **Trinity Grade 2 Piano** level